

# Brady Bangasser

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Software engineer across the stack, from low-latency Go and Rust services and secure backends to the infrastructure and CI that ship them.

## EXPERIENCE

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### System Administrator, Iowa State University

Apr 2025 – Present

Ames, IA

- Built a TUI stress-testing tool in C that dynamically schedules HPL (LINPACK) jobs and watches live metrics to keep runs inside safe thermal and power limits, protecting node health while pushing for peak results.
- Implemented ResMemPerGPU in a customized SLURM scheduler (C) to stop CPU jobs from consuming all memory on GPU nodes, protecting GPU workloads, with Makefile-driven builds and validation.

### Head of Networking, Cardinal Space Mining

Aug 2024 – Present

Ames, IA

- Collected multiple camera streams and relayed them to a third-party platform for live judging.
- Designed, built, and tested the internal network for a competition mining robot, ensuring redundant control and telemetry throughout each arena run.

### Cloud Security Engineer, John Deere

May 2025 – Aug 2025

Johnston, IA

- Led a migration of IAM, security, and dev-management tooling from Python to Go, a ~10x performance gain that cut AWS Lambda runtime costs by roughly the same factor.
- Deployed high-throughput Go services on AWS EKS to run compliance checks across the cloud estate.

### Research Assistant – Machine Learning, Iowa State University

Oct 2024 – May 2025

Ames, IA

- Designed and trained object-classification models with PyTorch and OpenCV, tuning data preprocessing and tensor operations to cut training latency without losing accuracy.
- Probed model robustness by building adversarial exploit methods, then designed defenses to harden the models against them.

### Lead Telemetry Engineer, Cyclone Rocketry

Aug 2024 – May 2025

Ames, IA

- Built modular software to control and acquire data from airframe instruments, improving reliability and maintainability.
- Built a ground-station TUI in C to visualize live telemetry from the airframe during flights.

## PROJECTS

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### Omni HTTP Router

2025–2026

- Built a Rust HTTP router that resolves middleware at compile time into fully static stacks, routing a request in under 30 instructions on SIMD-enabled x86.
- Supports gRPC and REST out of the box.

Tech: Rust, LLVM, Terraform, gRPC

### Minnesotan Programming Language

2023–2025

- Designed a Turing-complete esoteric language and built an end-to-end compiler with Flex/Bison that emits LLVM IR, compiling to native executables.
- Implemented a custom lexer and parser that transform the language's colloquial syntax into an AST before LLVM IR emission.

Tech: C, C++, Rust, Lua, Flex/Bison, LLVM

### Computer Vision Parking Enforcement Radar

2025–2026

- Built a real-time computer-vision service that detects parking-enforcement vehicles at ~95% accuracy and broadcasts locations to a user-facing app.
- Stored and queried detected-vehicle geolocation in PostgreSQL for fast spatial lookups.
- Architected a Redis Streams pipeline for low-latency, high-throughput image and detection routing from a REST API to processing servers, backed by PostgreSQL geospatial storage.

Tech: Python, Redis Streams, PostgreSQL, REST

### Public Data Scraping and Processing

2025

- Wrote a compiler-like tool that pairs lexical analysis with an LLM to standardize and validate data formats, and trained an ML model to flag anomalies in the collected data.
- Rendered target sites with a custom engine and generated Lua/Python API wrappers to collect data reliably.

Tech: C++, C, Python, Lua, Rust, MySQL, Docker, PyTorch, Terraform, Kubernetes

### PortTorpedo

2025

- Built a decentralized, end-to-end encrypted online Battleship using UDP hole punching for peer-to-peer play, with cryptographic compression and integrity checks.
- Implemented UDP hole punching so peers connect directly with no central server (TCP was explored but proved impractical to punch reliably).
- Added cryptographic integrity checks and compression over the peer-to-peer channel.

Tech: C++, C, AES, ECC, SHA3

### Gin Gonic Routing & Middleware Framework

2025–2026

- Wrote a Rust tool that auto-generates and protects Go (Gin) routes by file path, with middleware for permission bits, request IDs, and structured logging.
- Generated per-route middleware for permission bits, request IDs, and structured logging.

Tech: Rust, Go, Gin Gonic

### Secure Backend Architecture & Authentication

2025–2026

- Built a secure backend with user authentication and JWT sessions over a fully encrypted MySQL database, exposing scalable, high-throughput API routes.
- Exposed scalable, high-throughput API routes with authentication enforced on every endpoint.

Tech: Go, MySQL, JWT

### Lethal Company Sync

2023

- Built a multiplatform tool to sync files across devices using encryption, hashing, and compression.
- Synced only changed files using content hashing, and compressed transfers to cut bandwidth.

Tech: C++, C, Gzip, AES, Bcrypt, RSA, SHA3

## EDUCATION

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### M.S. Computer Science, Iowa State University

Aug 2025 – May 2027

GPA: 4.00

- Concurrent B.S./M.S. program

### B.S. Computer Science, Iowa State University

Aug 2024 – Dec 2026

GPA: 3.79 · Dean's List · Minors in Cybersecurity and Data Science

- Coursework: Compiler Design and Development, AI and Machine Learning in High Performance Computing, Network Architecture and Security, Theory of Computing, Software Design Practices

## TECHNICAL SKILLS

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**Languages:** C, C++, Python, Go, Rust, Bash, TypeScript, Java, SQL

**Infrastructure & Platform:** Docker, Linux/Unix, Git, CI/CD

**Data & Backends:** MySQL, PostgreSQL, Redis

**Security:** AES / RSA / ECDH, JWT / auth